# AN UNCONDITIONAL DENSITY BOUND ON ERDŐS–STRAUS EXCEPTIONAL PRIMES VIA THE SHARPNESS OF GATEWAY $A = 7$

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ABSTRACT. Let $E(X)$ denote the number of primes $p \leq X$ for which the Erdős–Straus equation $4/p = 1/x + 1/y + 1/z$ has no solution in positive integers and which are not covered by any of the known algebraic or gateway decompositions in the framework of [1]. We prove the first unconditional quantitative bound:

$$E(X) \;=\; O\left(\frac{X}{(\log X)^{3/2}}\right),$$

improving the trivial bound $O(X/\log X)$ by a factor of $(\log X)^{1/2}$. The proof uses the *sharpness* of gateway $A = 7$ (every Case B QR7 prime with an NQR prime factor in $N_7 = (p + 7)/4$ is solved by $A = 7$, proved unconditionally in [2]) to recast the exceptional set as a shifted-prime sequence sifted by the quadratic non-residues modulo 7, a sieve problem of dimension $\kappa = \frac{1}{2}$. A half-dimensional Selberg sieve with a Bombieri–Vinogradov level of distribution then gives the bound; only an upper bound is sought, so the parity obstruction does not intervene. We also characterise exactly which $A$ values can be "sharp" in this sense, confirm that $A = 7$ is the unique sharp gateway, and describe the character-sum barrier that prevents extending the method to give $E(X) = O(1)$.

## 1. Introduction

The Erdős–Straus conjecture (ES) asserts that for every integer $n \geq 2$ there exist positive integers $x, y, z$ satisfying

$$\frac{4}{n} \;=\; \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

It suffices to verify the conjecture for prime values of $n$. Write $E(X)$ for the number of prime counter-examples up to $X$. Elementary congruence arguments [1] reduce the problem to the *Case B QR7* residue class: primes $p \equiv 1 \pmod{24}$ with $p \equiv 1, 2,$ or $4 \pmod 7$ for which the fraction $(p + 3)/4$ is $Q_3$-smooth (all prime factors $\equiv 1 \pmod 3$). These primes constitute approximately 2.4% of all primes, but they resist the standard algebraic decompositions.

The *gateway decomposition framework* [1] handles each Case B QR7 prime using candidate parameters $A$, primes $\equiv 3 \pmod 4$, tried in increasing order until one succeeds. An initial implementation fixed a list of 28 values $A \in \{3, 7, 11, \ldots, 239\}$ and confirmed that one of them always succeeds for all primes $p \leq 10^9$ (with $A = 239$ the worst case); Section 4 analyses exactly this 28-value list. Section 6 reports the corrected, baseline-validated verification of all 4,118,054,813 primes to $10^{11}$ (superseding an earlier run with an inverted Case A/Case B classifier, see Remark 6.1), searching with no fixed candidate cap: only 32 distinct values of $A$ are ever required, the largest being $A = 359$ at $p = 3{,}807{,}728{,}761$. Exactly 47 primes below $10^{11}$ require $A \geq 199$, and only five require $A \geq 251$; there are no open cases.

Theoretically, the only prior bound was the trivial

$$E(X) \;\leq\; \pi_{\mathrm{CaseB}}(X) \;=\; O\!\left(\frac{X}{\log X}\right),$$

where $\pi_{\mathrm{CaseB}}(X)$ counts all Case B QR7 primes up to $X$. Our main result is:

**Theorem 1.1** (Main)**.** $E(X) = O\big(X/(\log X)^{3/2}\big)$. *The implicit constant is effectively computable.*

The key ingredient is the exact characterisation of gateway $A = 7$ failure, proved in [2]:

**Theorem 1.2** (A=7 Sharpness, [2, Theorem 5.2])**.** *For every Case B QR7 prime $p$, gateway $A = 7$ succeeds if and only if $N_7 = (p + 7)/4$ has at least one prime factor that is a quadratic non-residue modulo* 7.

This iff characterisation is the crucial bridge: it allows us to translate gateway failure into a multiplicative smoothness condition, amenable to analytic methods.

*Remark* 1.3. Theorem 1.2 holds for $A = 7$ and *no other* candidate $A$ in the framework. Section 4 explains exactly why.

**Organisation.** Section 2 recalls notation and the gateway machinery. Section 3 proves Theorem 1.1. Section 4 characterises sharp gateways. Section 5 describes the character-sum approach and the barrier to unconditional finitude. Section 6 reports the verification results to $10^{11}$. Section 7 lists open problems.

## 2. Setup and Notation

We follow the notation of [1, 2] throughout.

For a prime $A \equiv 3 \pmod 4$, write $Q_A$ for the set of quadratic residues modulo $A$ (including 0):

$$Q_A \;=\; \{r \in \mathbb{Z}/A\mathbb{Z} : r \equiv x^2 \pmod A \text{ for some } x\}.$$

The non-trivial quadratic residues form a subgroup of $(\mathbb{Z}/A\mathbb{Z})^*$ of index 2 and order $(A-1)/2$.

**Definition 2.1** ($Q_A$-smooth)**.** A positive integer $n$ is *$Q_A$-smooth* if every prime factor $q$ of $n$ satisfies $\left(\frac{q}{A}\right) = +1$.

For a Case B QR7 prime $p$ and a candidate parameter $A \equiv 3 \pmod 4$ with $4 \mid (p + A)$, set $N_A = (p + A)/4$. The *gateway target* is

$$t_A \;=\; -4N_A^2 \bmod A,$$

and the *gateway set* is

$$S_A(p) \;=\; \big\{d \bmod A : d \mid N_A^2,\ \gcd(d, Ap) = 1\big\}.$$

Gateway $A$ *succeeds* for prime $p$ if $t_A \in S_A(p)$.

The key lemma [1, Lemma 3.1] asserts:

**Lemma 2.2** (NQR Target)**.** *For every prime $A \equiv 3 \pmod 4$, the target $t_A$ is a quadratic non-residue modulo $A$.*

This follows immediately from $(-1/A) = -1$ when $A \equiv 3 \pmod 4$.

**Theorem 2.3** (Failure Characterisation, [2, Theorem 3.2])**.**
- (i) *If $N_A$ is $Q_A$-smooth, then $S_A(p) \subseteq Q_A \cup \{0\}$, so $t_A \notin S_A(p)$ and gateway $A$ fails.*
- (ii) *Conversely, if gateway $A$ succeeds, then $N_A$ is not $Q_A$-smooth.*
- (iii) *Having an NQR prime factor is necessary but not sufficient for gateway success: $S_A(p)$ may contain NQR residues without containing the specific target $t_A$.*

From (i): $\{p : N_A \text{ is } Q_A\text{-smooth}\} \subseteq \{p : \text{gateway } A \text{ fails}\}$.
From (ii): $\{p : \text{gateway } A \text{ succeeds}\} \subseteq \{p : N_A \text{ is not } Q_A\text{-smooth}\}$.

## 3. The Unconditional Density Bound

### 3.1. Reduction to a smoothness count.

**Lemma 3.1.** *$E(X) \leq \#\{p \leq X : p$ Case B QR7, $N_7$ is $Q_7$-smooth$\}$.*

*Proof.* If $p$ is an exceptional prime ($E(X)$ counter-example), every gateway fails. In particular, gateway $A = 7$ fails. By Theorem 1.2, the exact characterisation for $A = 7$, gateway $A = 7$ fails if and only if $N_7$ is $Q_7$-smooth. Hence every exceptional prime has $N_7$ $Q_7$-smooth. □

*Remark* 3.2. The inclusion is tight: if a Case B QR7 prime $p$ has $N_7$ $Q_7$-smooth, then gateway $A = 7$ certainly fails, but some other gateway $A \neq 7$ might still succeed for $p$. Thus the right-hand side of Lemma 3.1 is in general strictly larger than $E(X)$.

### 3.2. **Heuristic motivation: the density of $Q_7$-smooth integers.**

**Lemma 3.3** (Landau–Selberg for Chebotarev sets)**.** *Let $\mathcal{P}$ be a set of primes with Dirichlet (Chebotarev) density $\delta \in (0, 1)$. Call $n$ $\mathcal{P}$-*smooth *if every prime factor of $n$ lies in $\mathcal{P}$. Then*

$$\#\{n \le X : n \text{ is } \mathcal{P}\text{-smooth}\} \;\sim\; C_{\mathcal{P}} \frac{X}{(\log X)^{1-\delta}}$$

*for an explicit positive constant $C_{\mathcal{P}}$.*

This is a classical consequence of the Selberg–Delange method applied to the Dirichlet series $\sum_{n \mathcal{P}\text{-smooth}} n^{-s} = \prod_{q \in \mathcal{P}} (1 - q^{-s})^{-1}$, which has a singularity of order $\delta$ at $s = 1$; see [5, Chapter II.5] for the full statement.

For $Q_7$-smooth integers: the set of QR primes mod 7 has Chebotarev density $\delta = 1/2$ (half of all primes, by quadratic reciprocity and Dirichlet's theorem). Lemma 3.3 gives:

$$\#\{n \le X : n \text{ is } Q_7\text{-smooth}\} \;\sim\; C_7 \frac{X}{(\log X)^{1/2}}. \tag{1}$$

If the values $N_7 = (p+7)/4$ behaved, with respect to $Q_7$-smoothness, like random integers of size $X/4$, then by (1) a proportion $\asymp (\log X)^{-1/2}$ of the $\asymp X/\log X$ admissible primes would have $N_7$ smooth, predicting $E(X) \asymp X/(\log X)^{3/2}$. We make this rigorous as an *upper* bound. The integer $N_7$ is a shifted prime, not a random integer, so the heuristic must be replaced by a sieve. Because only an upper bound is required, the parity obstruction does not arise, and a half-dimensional sieve with a Bombieri–Vinogradov level of distribution suffices.

### 3.3. **The exceptional set as a sifted sequence.**

Fix the three residue classes modulo $168 = \mathrm{lcm}(24, 7)$ determined by $p \equiv 1 \pmod{24}$ and $p \equiv 1, 2, 4 \pmod 7$; write $\mathcal{R} \subseteq (\mathbb{Z}/168\mathbb{Z})^*$ for this set, so $|\mathcal{R}| = 3$. Define the sifting sequence

$$\mathcal{A} \;=\; \Big\{ (p+7)/4 \;:\; p \le X \text{ prime},\; p \bmod 168 \in \mathcal{R} \Big\},$$

and the sifting set of primes (the non-residues modulo 7)

$$\mathcal{P} \;=\; \big\{ \ell \text{ prime} : \ell \equiv 3, 5, 6 \pmod 7 \big\}.$$

Write $P(z) = \prod_{\ell \in \mathcal{P},\, \ell < z} \ell$ and

$$S(\mathcal{A}, \mathcal{P}, z) \;=\; \#\{\, a \in \mathcal{A} : \gcd(a, P(z)) = 1 \,\}.$$

**Lemma 3.4** (Smoothness forces survival)**.** *For every $z \geq 2$,*

$$\#\{p \leq X : p \bmod 168 \in \mathcal{R},\ N_7 \textit{ is } Q_7\textit{-smooth}\} \;\leq\; S(\mathcal{A}, \mathcal{P}, z).$$

*Proof.* If $N_7 = (p+7)/4$ is $Q_7$-smooth then every prime factor of $N_7$ is a quadratic residue modulo 7, so $N_7$ has no prime factor in $\mathcal{P}$; in particular $\gcd(N_7, P(z)) = 1$, and $p$ contributes the element $a = N_7$ to the survivors. Here we use that for odd $\ell \neq 7$, $\ell \mid (p+7)/4 \iff \ell \mid (p+7)$ since $\gcd(\ell, 4) = 1$; every $\ell \in \mathcal{P}$ is such, because $2, 7 \notin \mathcal{P}$ (indeed $2 \equiv 3^2$ is a residue and $7 \equiv 0 \pmod 7$). □

Combining Lemmas 3.1 and 3.4, $E(X) \leq S(\mathcal{A}, \mathcal{P}, z)$ for every $z \geq 2$.

### 3.4. **Local densities and the sieve dimension.**

**Lemma 3.5.** *For squarefree $d \mid P(z)$ write $\mathcal{A}_d = \{a \in \mathcal{A} : d \mid a\}$. Then* $\gcd(d, 168) = 1$ *and*

$$|\mathcal{A}_d| \;=\; \frac{\omega(d)}{d} X_0 \;+\; r_d, \qquad \frac{\omega(d)}{d} = \frac{1}{\phi(d)}, \qquad X_0 = \frac{3}{\phi(168)} \operatorname{Li}(X),$$

*where $r_d = \sum_{i=1}^{3} \big(\pi(X; 168d, c_i) - \operatorname{Li}(X)/\phi(168d)\big)$ for residues $c_i$ supplied by the Chinese Remainder Theorem. The density $\omega$ is multiplicative with $\omega(\ell)/\ell = 1/(\ell - 1)$ for $\ell \in \mathcal{P}$, and the sieve has dimension $\kappa = \frac{1}{2}$:*

$$V(z) \;:=\; \prod_{\substack{\ell < z \\ \ell \in \mathcal{P}}} \Big(1 - \frac{\omega(\ell)}{\ell}\Big) \;=\; \prod_{\substack{\ell < z \\ \ell \in \mathcal{P}}} \Big(1 - \frac{1}{\ell - 1}\Big) \;\asymp\; (\log z)^{-1/2}.$$

*Proof.* For $\ell \in \mathcal{P}$ we have $\ell \nmid 168$, and as in Lemma 3.4 $\ell \mid a = (p+7)/4 \iff p \equiv -7 \pmod{\ell}$. Since $\gcd(-7, \ell) = 1$ this is one class modulo $\ell$, coprime to $\ell$. By the Chinese Remainder Theorem the conditions "$p \bmod 168 \in \mathcal{R}$ and $p \equiv -7 \pmod d$" select exactly 3 classes modulo $168d$, all coprime to $168d$ (using $\gcd(d, 168) = 1$). Counting primes in these classes gives $|\mathcal{A}_d|$ with main term $3 \operatorname{Li}(X)/\phi(168d) = \frac{1}{\phi(d)} \cdot \frac{3 \operatorname{Li}(X)}{\phi(168)}$, i.e. $\omega(d)/d = 1/\phi(d)$, multiplicative with $\omega(\ell)/\ell = 1/(\ell - 1)$.

For the dimension, $\log(1 - \frac{1}{\ell - 1}) = -\frac{1}{\ell} + O(\ell^{-2})$, so

$$-\log V(z) = \sum_{\substack{\ell < z \\ \ell \in \mathcal{P}}} \frac{1}{\ell} + O(1) = \tfrac{1}{2} \log\log z + O(1),$$

by Mertens' theorem for arithmetic progressions: the residues $3, 5, 6 \pmod 7$ comprise three of the $\phi(7) = 6$ reduced classes, each contributing $\frac{1}{6}\log\log z + O(1)$ to $\sum_{\ell<z} 1/\ell$. Exponentiating gives $V(z) \asymp (\log z)^{-1/2}$. ☐

3.5. **Level of distribution.**

**Lemma 3.6** (Bombieri–Vinogradov input)**.** *For every $K > 0$, with $D = X^{2/5}$,*

$$\sum_{\substack{d\le D\\ d\mid P(z)}} \mu^2(d)\, 3^{\nu(d)}\, |r_d| \ \ll_K \ \frac{X}{(\log X)^K}.$$

*Proof.* By the triangle inequality, writing $\Delta(q, a) = \pi(X; q, a) - \mathrm{Li}(X)/\phi(q)$,

$$|r_d| \ \le \ \sum_{i=1}^{3} |\Delta(168d, c_i)| \ \le \ 3 \max_{(a,168d)=1} |\Delta(168d, a)|,$$

and $168d \le 168D \le X^{1/2}$ for large $X$. Cauchy–Schwarz gives

$$\sum_{d\le D} 3^{\nu(d)}|r_d| \le \Big(\sum_{d\le D} \frac{9^{\nu(d)}}{\phi(d)} \cdot X\log X\Big)^{1/2} \Big(\sum_{d\le D} |r_d|\Big)^{1/2},$$

using the trivial bound $|r_d| \ll X\log X/\phi(d)$. The first factor is $\ll X^{1/2}(\log X)^5$ since $\sum_{d\le D} 9^{\nu(d)}/\phi(d) \ll (\log X)^9$; the second is $\ll \big(X/(\log X)^{K'}\big)^{1/2}$ for any $K'$ by the Bombieri–Vinogradov theorem [4, Thm. 17.1], valid since $168D \le X^{1/2}$. Choosing $K'$ large gives the claim. ☐

3.6. **Proof of Theorem 1.1.** Take $z = X^{1/5}$, so the Selberg upper-bound sieve with weights supported on $d \le \xi = z$ has remainder over moduli $d \le D = \xi^2 = X^{2/5}$. By Lemma 3.5 the sequence $\mathcal{A}$ satisfies the $\kappa$-dimensional hypothesis of Selberg's sieve with $\kappa = \frac{1}{2}$ and density $\omega$ (note $\omega(\ell)/\ell = 1/(\ell-1) \le \frac{1}{2}$ for $\ell \ge 3$), and by Lemma 3.6 it has level of distribution $D = X^{2/5} \le X^{1/2-\epsilon}$, within Bombieri–Vinogradov range. Selberg's sieve [3, Thm. 2.2 and the $\kappa$-dimensional bound of §2.4], [4, Ch. 6] yields

$$S(\mathcal{A}, \mathcal{P}, z) \ \ll \ X_0\, V(z) \ + \ \sum_{\substack{d\le D\\ d\mid P(z)}} \mu^2(d)\, 3^{\nu(d)}\, |r_d|.$$

By Lemma 3.5, $V(z) \asymp (\log z)^{-1/2} \asymp (\log X)^{-1/2}$ since $\log z = \frac{1}{5}\log X$, and $X_0 \asymp X/\log X$; the remainder is $\ll_K X/(\log X)^K$ for any $K$ by Lemma 3.6. Hence

$$S(\mathcal{A}, \mathcal{P}, z) \ \ll \ \frac{X}{\log X} \cdot \frac{1}{(\log X)^{1/2}} \ = \ \frac{X}{(\log X)^{3/2}}.$$

By Lemmas 3.1 and 3.4, $E(X) \le S(\mathcal{A}, \mathcal{P}, z)$, proving the theorem. The implied constant is effective, as the Bombieri–Vinogradov theorem and the Selberg bound are effective for the moduli involved. □

*Remark* 3.7 (Only an upper bound is claimed). The argument bounds $E(X)$ from above and uses no asymptotic for $S(\mathcal{A}, \mathcal{P}, z)$. A matching lower bound, or an asymptotic for $\#\{p \le X : N_7 \text{ is } Q_7\text{-smooth}\}$, would meet the parity obstruction of sieve theory and is not needed here. The exponent $3/2$ is exactly $1 + \kappa$ with $\kappa = \frac{1}{2}$ the relative density of non-residues modulo 7; the saving $(\log X)^{-1/2}$ is intrinsic to the quadratic splitting at $A = 7$.

## 4. Sharp Gateways and the Uniqueness of $A = 7$

**Definition 4.1** (Sharp gateway)**.** A candidate parameter $A \equiv 3 \pmod 4$ is *sharp* if, for every Case B QR7 prime $p$ for which $4 \mid (p + A)$, gateway $A$ succeeds *if and only if* $N_A$ is not $Q_A$-smooth; and moreover the characterisation is *non-vacuous*: there exist Case B QR7 primes $p$ for which $N_A$ is not $Q_A$-smooth (so gateway $A$ sometimes succeeds on Case B QR7 primes).

*Remark* 4.2. The non-vacuity clause excludes $A = 3$. By definition, Case B QR7 primes satisfy $(p + 3)/4$ is $Q_3$-smooth, so $N_3 = (p + 3)/4$ is always $Q_3$-smooth and gateway $A = 3$ always fails on Case B QR7 primes. The iff holds vacuously but gives no density improvement.

By Theorem 2.3(i)(ii), every gateway satisfies the forward implication ($Q_A$-smooth ⇒ fails) and the converse direction at the level of necessary conditions (success ⇒ not smooth). A sharp gateway additionally satisfies the reverse: not smooth ⇒ succeeds.

The proof of Theorem 1.2 in [2] proceeds by:

(1) Identifying the *forced* QR factor $g = 2$ of $N_7$: since $p \equiv 1 \pmod 8$, we have $8 \mid (p + 7)$, hence $2 \mid N_7$.
(2) Observing that 2 *generates* $Q_7 = \{1, 2, 4\}$: the powers $2^0, 2^1, 2^2$ cycle through all of $Q_7 \pmod 7$.
(3) Showing that if $q \mid N_7$ with $\left(\frac{q}{7}\right) = -1$, then $q \cdot 2^j \bmod 7$ hits every NQR modulo 7 as $j \in \{0, 1, 2\}$, including the target $t_7$.
(4) Verifying the integrality condition: $d = q \cdot 2^{2k}$ divides $N_7^2$ (since $v_2(N_7^2) = 2v_2(N_7) \ge 2$), and $\gcd(d, 7p) = 1$ by Lemma 3.2 of [1].

**Theorem 4.3** (Uniqueness of $A = 7$)**.** *Among the* 28 *candidate $A$ values in the framework, $A = 7$ is the only sharp gateway.*

*Proof.* Sharpness requires a prime $g$ that is *forced* to divide $N_A$ for every Case B QR7 prime with $4 \mid (p + A)$, such that (a) $g$ is a QR modulo $A$, (b) $g$ generates $Q_A$ under multiplication modulo $A$, and (c) the guaranteed valuation suffices: $2v_g(N_A) \geq |Q_A| - 1$, so that $\{g^0, g^1, \ldots, g^{2v_g}\}$ realises all of $Q_A$ as divisor residues of $N_A^2$.

*Which primes can be forced.* Case B QR7 fixes $p \equiv 1 \pmod 8$ and $p \equiv 1 \pmod 3$, but leaves $p \bmod \ell$ free for every prime $\ell \geq 5$. Hence the only primes that can divide $N_A = (p + A)/4$ for *all* such $p$ are 2 and 3, and a direct check of $4N_A = p + A$ gives the rule

$$2 \mid N_A \iff A \equiv 7 \pmod 8, \qquad 3 \mid N_A \iff A \equiv 2 \pmod 3.$$

A candidate with neither $A \equiv 7 \pmod 8$ nor $A \equiv 2 \pmod 3$ has no forced prime factor and cannot be sharp.

We check each candidate:

$A = 7$**:** $7 \equiv 7 \pmod 8$, so $g = 2$ is forced; $7 \equiv 1 \pmod 3$, so 3 is not. $(2/7) = +1$, and $\mathrm{ord}_7(2) = 3 = |Q_7|$, so 2 generates $Q_7$. The guaranteed $v_2(N_7) \geq 1$ gives $2v_2(N_7) \geq 2 = |Q_7| - 1$. **Sharp.**

$A = 11$**:** $11 \equiv 2 \pmod 3$, so $g = 3$ is forced $(3 \mid N_{11})$; $11 \equiv 3 \pmod 8$, so 2 is not. $(3/11) = +1$ (QR, since $3 \in \{1, 3, 4, 5, 9\}$). Order of 3 in $(\mathbb{Z}/11\mathbb{Z})^*$ is $5 = |Q_{11}|$, so 3 generates $Q_{11}$. However, the minimum forced multiplicity is $v_3(N_{11}) \geq 1$, giving $2v_3(N_{11}) \geq 2$, but we need $\geq |Q_{11}| - 1 = 4$. For $p \equiv 1, 4 \pmod 9$ (two-thirds of cases) one has $v_3(N_{11}) = 1$, so the required divisors $3^3$ and $3^4$ do not appear in $N_{11}^2$ and the target $t_{11}$ need not be hit.

*Concrete counterexample.* Take $p = 193$ (Case B QR7: $193 \equiv 1 \pmod{24}$, $193 \equiv 4 \pmod 7$, $(p+3)/4 = 49 = 7^2$ is $Q_3$-smooth). Then $N_{11} = (193+11)/4 = 51 = 3 \times 17$. The factor 17 satisfies $(17/11) = -1$ (NQR: $6 \notin Q_{11}$), so $N_{11}$ has an NQR prime factor. Yet $t_{11} = (-4 \cdot 51^2) \bmod 11 = 2$, while $S_{11}(193) = \{1, 3, 5, 6, 7, 9, 10\}$ (all residues of divisors of $51^2 = 3^2 \cdot 17^2$ modulo 11), and $2 \notin S_{11}(193)$. Gateway $A = 11$ fails for $p = 193$ despite $N_{11}$ having an NQR factor. **Not sharp.**

$A = 19$**:** $19 \equiv 3 \pmod 8$ and $19 \equiv 1 \pmod 3$, so *neither* 2 nor 3 is forced into $N_{19}$. No prime is forced, so condition for sharpness fails at the first step. (For instance $p = 193$ is Case B QR7 with $N_{19} = 53$, prime and coprime to 2 and 3.) **Not sharp.**

$A \geq 23$ **(remaining 25 candidates):** A systematic check of all 28 candidate values confirms that for every $A \in \{23, 31, 43, \ldots, 239\}$, either (a) no forced prime ($A \not\equiv 7 \pmod 8$ and $A \not\equiv 2 \pmod 3$, e.g. $A = 43, 67, 79, \ldots$), or (b) a forced prime exists but the guaranteed valuation satisfies $2v_g(N_A) < |Q_A| - 1$. For $A = 23$ ($\equiv 7 \pmod 8$ and $\equiv 2 \pmod 3$) both 2 and 3 are forced, but $|Q_{23}| = 11$ requires $2v_g \geq 10$ while the guaranteed valuations are 1 each. The growing gap $|Q_A| - 1 \geq 10$ against bounded forced valuations rules out every $A \geq 23$. **None sharp.**

The computation was verified by testing all 28 candidates directly, confirming $A = 7$ is the unique potentially sharp gateway. □

*Remark* 4.4. The special arithmetic of $A = 7$: $|Q_7| = 3$ is the smallest possible, and $|Q_7| - 1 = 2$ equals $2 \cdot \min(v_2(N_7)) = 2 \cdot 1$. No larger prime $A$ can match this coincidence.

## 5. Character Sums and the Barrier to Finitude

The density bound of Theorem 1.1 does not imply finiteness of $E$: $X/(\log X)^{3/2} \to \infty$ as $X \to \infty$. We describe the character-sum approach that would be needed to obtain $E(X) = O(1)$.

5.1. **Character sum formula for gateway solutions.** The number of gateway solutions for parameter $A$ is:

$$\mathcal{N}_A(p) \;=\; \frac{1}{\varphi(A)} \sum_{\chi \bmod A} \bar{\chi}(t_A) \sum_{\substack{d \mid N_A^2 \\ \gcd(d,Ap)=1}} \chi(d). \tag{2}$$

Separating the trivial character $\chi_0$ and the Legendre symbol $\chi_2$:

$$\mathcal{N}_A(p) \;=\; \frac{\tau'(N_A^2) - \tau'(M_A^2)}{\varphi(A)} \;+\; E_A(p), \tag{3}$$

where $\tau'(n)$ counts the divisors of $n$ coprime to $A$, $M_A$ is the $Q_A$-smooth part of $N_A$, and $E_A(p)$ involves only characters of order $\geq 3$.

*Remark* 5.1. The restriction in $\tau'$ is inherited from $\gcd(d, Ap) = 1$ in (2) and is vacuous here: $4N_A = p + A$, so $A \mid N_A$ would force $A \mid p$, impossible for $A < p$ prime. Hence $\tau'(N_A^2) = \tau(N_A^2)$ throughout, and likewise for $M_A$. The prime is kept because the identity $\sum_{d \mid N_A^2,\,(d,A)=1} \chi_2(d) \;=\; \tau'(M_A^2)$, which produces the second term, is stated for the coprime-restricted sum.

The *main term* $\tau'(N_A^2) - \tau'(M_A^2)$ is positive if and only if $N_A$ is not $Q_A$-smooth (i.e., $M_A < N_A$). Gateway $A$ fails only when $\mathcal{N}_A(p) = 0$, which requires $E_A(p) \leq -(\tau'(N_A^2) - \tau'(M_A^2))/\varphi(A)$.

5.2. **Why the character sum barrier is hard.** For $A = 7$: $\varphi(7) = 6$ has characters of orders $1, 2, 3, 6$. The order-3 characters are the "next-order" corrections. Bounding $|E_7(p)|$ requires nontrivial estimates for twisted divisor sums $\sum_{d \mid N_7^2} \chi_3(d)$.

For general $A$: bounding $\sum_{p \leq X} |\mathcal{N}_A(p)|^{-\epsilon}$ (which would suffice for finitude under GRH-type hypotheses) is an open problem.

5.3. **What additional hypotheses give.**

**Theorem 5.2** (Conditional, [2, Theorem 7.2])**.** *Under Conjecture 7.1 of* [2] *(joint $Q_A$-smoothness equidistribution, weaker than Elliott–Halberstam), the Erdős–Straus conjecture holds for all sufficiently large primes.*

The gap between Theorem 1.1 and finiteness corresponds to the gap between:

- $O(X/(\log X)^{3/2})$ (what smoothness density gives), and
- $O(1)$ (what finiteness requires).

To close this gap unconditionally, one would need to show that the character sum error $E_A(p)$ is never large enough to cancel the main term across all 28 gateways simultaneously.

5.4. **Why more gateways do not trivially improve the exponent.** It is tempting to improve Theorem 1.1 by using all 28 candidates at once. The naive heuristic runs as follows. For primes $q > 239$ each $q$ divides at most one $N_{A_i}$ (since $N_{A_i} - N_{A_j} = (A_i - A_j)/4 \leq 58 < q$ in the 28-candidate set), and by Chebotarev the Frobenius class of $q$ at each gateway $A_i$ is independent. The probability that every large prime factor of $N_{A_i}$ is QR mod $A_i$ simultaneously for $i = 1, \ldots, k$ would then decay as $(\log X)^{-k/2}$, suggesting $E(X) = O(X/(\log X)^{1+k/2})$.

This heuristic is *not rigorous*, and the obstruction is structural. The bound of Theorem 1.1 uses the *sharpness* of $A = 7$: the exceptional set is contained in $\{p : N_7 \text{ is } Q_7\text{-smooth}\}$ because for $A = 7$, gateway failure is *equivalent* to $Q_7$-smoothness. For every other $A$ the equivalence fails (Theorem 2.3(iii)): a prime can fail gateway $A_i$ while $N_{A_i}$ is *not* $Q_{A_i}$-smooth. Consequently the exceptional set is *not* contained in the joint-smoothness set $\bigcap_i \{N_{A_i} \text{ is } Q_{A_i}\text{-smooth}\}$, and bounding the latter by a Selberg sieve gives *no* upper bound on $E(X)$. Indeed $\bigcap_i \{N_{A_i} \text{ smooth}\} \subseteq E$ runs the wrong way, a lower bound on $E$ if anything.

A genuine improvement therefore requires controlling the *character-sum* failure events $\{\mathcal{N}_{A_i}(p) = 0\}$ of (2) for $A_i \neq 7$, not merely smoothness. We record the expected truth as a conjecture, with the caveat that it is not a sieve statement:

**Conjecture 5.3.** $E(X) = O\big(X/(\log X)^{1+\delta}\big)$ for every $\delta > 0$; conjecturally $E(X) = O(1)$.

A proof would need joint control of the divisor character sums across the candidate gateways, the analytic gap described in [2]; the smoothness method of Theorem 1.1 is intrinsically limited to the single sharp gateway $A = 7$ and the exponent $3/2$.

## 6. Verification to $10^{11}$

6.1. **Method.** The reported numbers come from the corrected, baseline-validated pipeline (`sessions/session20_corrected_10B.py` in the software repository), which:

(1) generates primes with a CPU segmented sieve (segments of $10^8$);
(2) classifies each prime into the algebraic cases of [1] or the Case B QR7 residual class, using the corrected Case A/Case B test (Remark 6.1);
(3) for each Case B QR7 prime, searches candidate values $A \equiv 3 \pmod 4$ in increasing order — with no fixed upper cap, up to $A < 10^4$ — factorising $N_A$ and testing the divisor-residue condition of Section 2, recording the first successful $A$;
(4) runs on a 16-worker CPU pool with resumable, per-segment checkpointing, validating against three independent baselines (the full $A$-distribution at $10^6$, the residual count at $10^7$, and the residual count and max $A$ at $10^9$) before extending the range.

An earlier GPU-accelerated implementation (CuPy, fixed candidate lists of 28 then 87 values) first located the $A = 359$ record described in Section 6.3; it has since been superseded by the pipeline above, which is the source of every number in this section.

*Remark* 6.1 (Data integrity). An earlier run of the $10^{10}$ stage (March 2026) inverted the Case A/Case B test, skipping every residual-class prime as already proven and instead gateway-searching the complementary (easy) class. The resulting figures — a residual count of 18,189,229 at $10^{10}$, $\max A = 251$, a 25-value $A$-dictionary — described the wrong set of primes and are withdrawn; they do not appear in this paper. The corrected pipeline described above reproduces the correct-classifier baselines at $10^6$, $10^7$, and $10^9$ before extending the range, and every reported solution is verified by direct evaluation of $4/p = 1/x+1/y+1/z$.

6.2. **Results to $10^{11}$.**

| Limit | Primes | Case B QR7 | Max $A$ required |
|---|---:|---:|---:|
| $10^6$ | 78,498 | 2,269 | 79 |
| $10^7$ | 664,579 | 18,019 | 167 |
| $10^8$ | 5,761,455 | 146,227 | 239 |
| $10^9$ | 50,847,534 | 1,212,383 | 239 |
| $10^{10}$ | 455,052,511 | 10,246,512 | **359** (see §6.3) |
| $10^{11}$ | 4,118,054,813 | 88,088,687 | **359** (unchanged from $10^{10}$) |

**Theorem 6.2** (Computational Verification)**.** *The Erdős–Straus conjecture holds for all primes $p \leq 10^{11}$, with maximum gateway parameter $A \leq 359$ and zero open cases. Exactly* 47 *primes below* $10^{11}$ *require $A \geq 199$ (*12 *of them below* $10^{10}$*), and only five require $A \geq 251$: $p = 3{,}807{,}728{,}761$ ($A = 359$), $p = 31{,}048{,}497{,}481$ ($A = 347$), $p = 6{,}372{,}847{,}849$ ($A = 311$), $p = 26{,}147{,}464{,}729$ ($A = 251$), and $p = 94{,}604{,}730{,}049$ ($A = 251$). The record $A = 359$ is attained below $10^{10}$ and is not exceeded anywhere in* $(10^{10}, 10^{11}]$.

6.3. **Anatomy of the hardest prime.** The unique prime below $10^{11}$ requiring $A = 359$ is

$$p \;=\; 3{,}807{,}728{,}761.$$

*Verification that $p$ is Case B QR7.* $p \equiv 1 \pmod{24}$, $p \equiv 4 \pmod 7$, and $(p+3)/4 = 951{,}932{,}191 = 7 \times 73 \times 1{,}862{,}881$, all three factors $\equiv 1 \pmod 3$, confirming $Q_3$-smoothness.

*Solution via $A = 359$.* $N_{359} = (p + 359)/4 = 951{,}932{,}280 = 2^3 \cdot 3 \cdot 5 \cdot 13 \cdot 23 \cdot 43 \cdot 617$. The factors $13, 43, 617$ are all NQR modulo 359. The divisor $d = 1935 = 3^2 \times 5 \times 43$ divides $N_{359}^2$, $\gcd(d, 359p) = 1$, and $1935 \bmod 359 = 140 = t_{359}$, so gateway $A = 359$ succeeds. This claim is independently reproduced by `verify_hardest_prime.py` in the software repository.

This prime is the record-holder over the full $10^{11}$ range: the largest gateway parameter required for any prime $p \leq 10^{11}$. Of the four other primes requiring $A \geq 251$ (Theorem 6.2), one — $p = 6{,}372{,}847{,}849$ ($A = 311$) — also lies below $10^{10}$, alongside the $A = 359$ record; the remaining three lie in $(10^{10}, 10^{11}]$. None exceeds $A = 359$.

*Remark* 6.3. The Case B QR7 residual class has relative density 2.14% at $10^{11}$ (down from 2.89% at $10^6$), consistent with the relative-density-zero prediction for this class in [1]. The fraction requiring large $A$ shrinks even faster: only 47 of the 88,088,687 Case B QR7 primes below $10^{11}$ need $A \geq 199$.

## 7. Open Problems

*Question* 7.1. Improve the bound to $E(X) = O(X/(\log X)^{3/2+\epsilon})$ for some $\epsilon > 0$ using properties of gateways beyond $A = 7$.

*Question* 7.2. Improve the exponent toward Conjecture 5.3 ($E(X) = O(X/(\log X)^{1+\delta})$ for all $\delta > 0$) by controlling the divisor character sums $\mathcal{N}_{A_i}(p)$ at gateways $A_i \neq 7$, not merely smoothness. As noted in §5, the joint $Q_A$-smoothness set does not contain the exceptional set, so the single-gateway smoothness method caps at the exponent 3/2.

*Question* 7.3. Extend the verification beyond $10^{11}$. The corrected pipeline (Section 6) confirms $A = 359$ as the hardest case for $p \leq 10^{11}$, unchanged since $10^{10}$; extending further would determine whether larger gateway parameters ever appear. The maximum $A$ encountered for any prime $p \leq X$ grows slowly with $X$ (log-like); bounding this growth rigorously is an open problem (the *bounded-A conjecture* of [1]).

*Question* 7.4. Characterise the set of primes $p$ for which $N_7$ is $Q_7$-smooth. By Theorem 1.1, this set has counting function $O(X/(\log X)^{3/2})$, but no explicit family is known. These are precisely the primes for which the A=7 gateway fails.

*Question* 7.5. Is there an $A \neq 7$ for which an exact characterisation of gateway failure holds for a *positive-density* subset of Case B QR7 primes? Theorem 4.3 rules out universal sharpness, but a density-one subset might still be achievable for specific $A$.

*Question* 7.6. Can the character sum formula (2) be combined with the large sieve to give an unconditional proof that $\mathcal{N}_A(p) > 0$ for all Case B QR7 primes $p \geq P_0$ (for explicit $P_0$)? This would require bounding $E_A(p)$ in terms of the main term and a zero-free region for $L(s, \chi)$ for characters $\chi$ mod $A$.

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This work uses techniques developed in [1, 2]. An initial GPU-accelerated search (CuPy, NVIDIA Quadro RTX 5000) first located the $A = 359$ record of Section 6.3; the results reported here come from the corrected CPU pipeline of Section 6.

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